

EDA-Practice

January 30, 2026

```
[1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: df = pd.read_csv(r"C:\Users\Madhu\Desktop\KMIT\AY-25-26-sem-2\Practical\train.
↪CSV")
df
```

```
[2]:
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	
..	...	...	...	...	
886	Montvila, Rev. Juozas	male	27.0	0	
887	Graham, Miss. Margaret Edith	female	19.0	0	
888	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	
889	Behr, Mr. Karl Howell	male	26.0	0	
890	Dooley, Mr. Patrick	male	32.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S

3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S
..	...	...	...	...	...
886	0	211536	13.0000	NaN	S
887	0	112053	30.0000	B42	S
888	2	W./C. 6607	23.4500	NaN	S
889	0	111369	30.0000	C148	C
890	0	370376	7.7500	NaN	Q

[891 rows x 12 columns]

```
[3]: df.shape
```

```
[3]: (891, 12)
```

```
[4]: df.head()
```

```
[4]: PassengerId  Survived  Pclass  \
0             1         0         3
1             2         1         1
2             3         1         3
3             4         1         1
4             5         0         3
```

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

```
[5]: df.sample(5)
```

```
[5]: PassengerId  Survived  Pclass  \
224           225         1         1
217           218         0         2
85            86         1         3
829           830         1         1
746           747         0         3
```

	Name	Sex	Age	SibSp	\
224	Hoyt, Mr. Frederick Maxfield	male	38.0	1	
217	Jacobsohn, Mr. Sidney Samuel	male	42.0	1	
85	Backstrom, Mrs. Karl Alfred (Maria Mathilda Gu...	female	33.0	3	
829	Stone, Mrs. George Nelson (Martha Evelyn)	female	62.0	0	
746	Abbott, Mr. Rossmore Edward	male	16.0	1	

	Parch	Ticket	Fare	Cabin	Embarked
224	0	19943	90.00	C93	S
217	0	243847	27.00	NaN	S
85	0	3101278	15.85	NaN	S
829	0	113572	80.00	B28	NaN
746	1	C.A. 2673	20.25	NaN	S

```
[6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
7   Parch           891 non-null    int64
8   Ticket          891 non-null    object
9   Fare            891 non-null    float64
10  Cabin           204 non-null    object
11  Embarked        889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
[7]: df.isnull().sum()
```

```
[7]: PassengerId     0
Survived           0
Pclass             0
Name               0
Sex                0
Age               177
SibSp              0
Parch              0
Ticket             0
Fare               0
```

```
Cabin          687
Embarked       2
dtype: int64
```

```
[8]: df.describe()
```

```
[8]:
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
[9]: df.duplicated().sum()
```

```
[9]: 0
```

```
[10]: df.corr(numeric_only=True)
```

```
[10]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch \
PassengerId	1.000000	-0.005007	-0.035144	0.036847	-0.057527	-0.001652
Survived	-0.005007	1.000000	-0.338481	-0.077221	-0.035322	0.081629
Pclass	-0.035144	-0.338481	1.000000	-0.369226	0.083081	0.018443
Age	0.036847	-0.077221	-0.369226	1.000000	-0.308247	-0.189119
SibSp	-0.057527	-0.035322	0.083081	-0.308247	1.000000	0.414838
Parch	-0.001652	0.081629	0.018443	-0.189119	0.414838	1.000000
Fare	0.012658	0.257307	-0.549500	0.096067	0.159651	0.216225

	Fare
PassengerId	0.012658
Survived	0.257307
Pclass	-0.549500
Age	0.096067
SibSp	0.159651

```
Parch      0.216225
Fare        1.000000
```

```
[11]: mean_age = df['Age'].mean()
      mean_age
```

```
[11]: 29.69911764705882
```

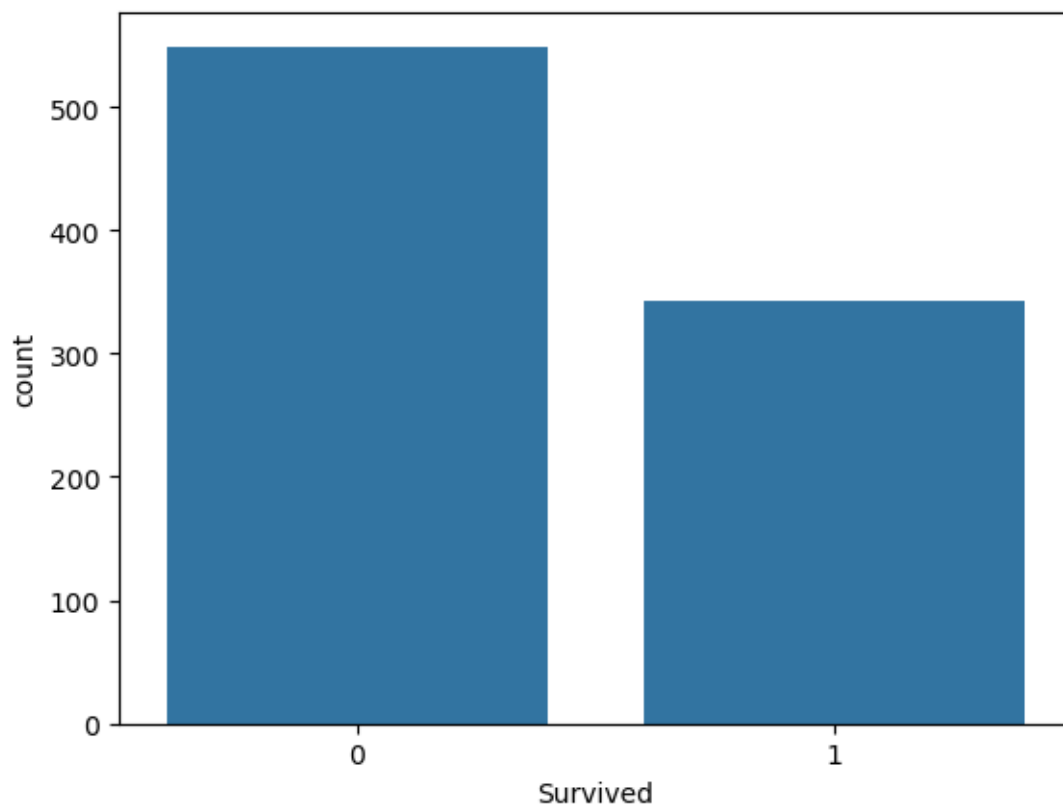
```
[12]: median_age = df['Age'].median()
      median_age
```

```
[12]: 28.0
```

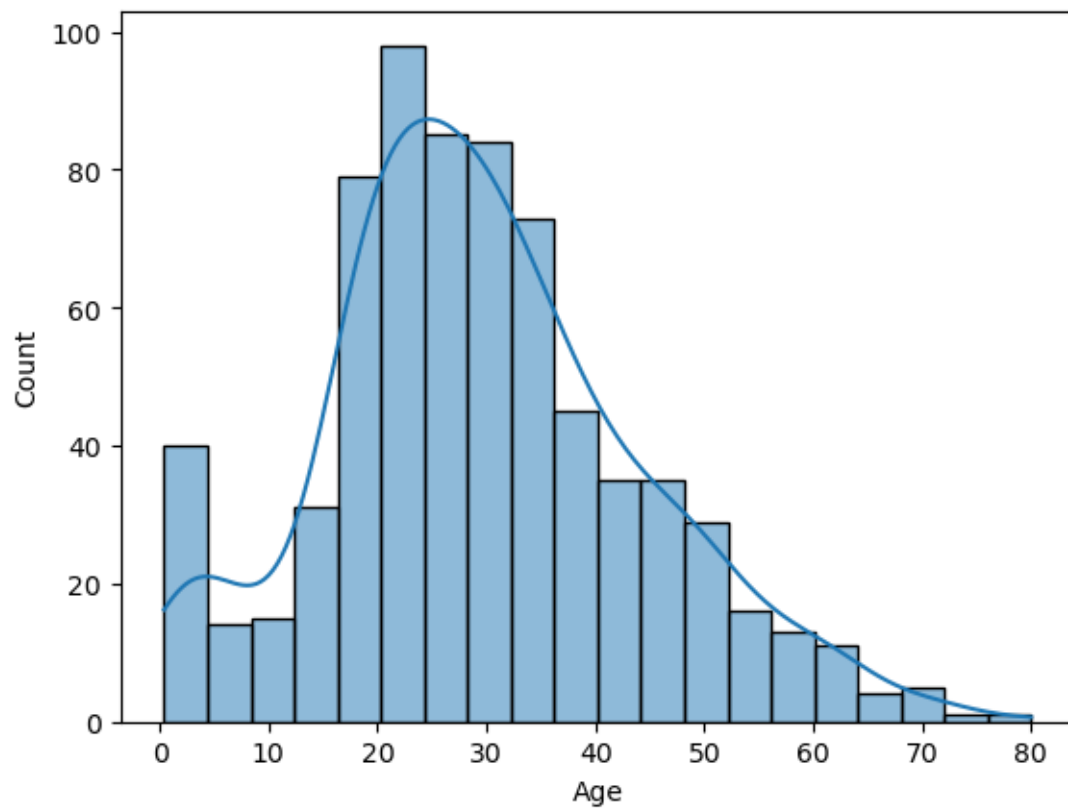
```
[13]: mode_age = df['Age'].mode()[0]
      mode_age
```

```
[13]: 24.0
```

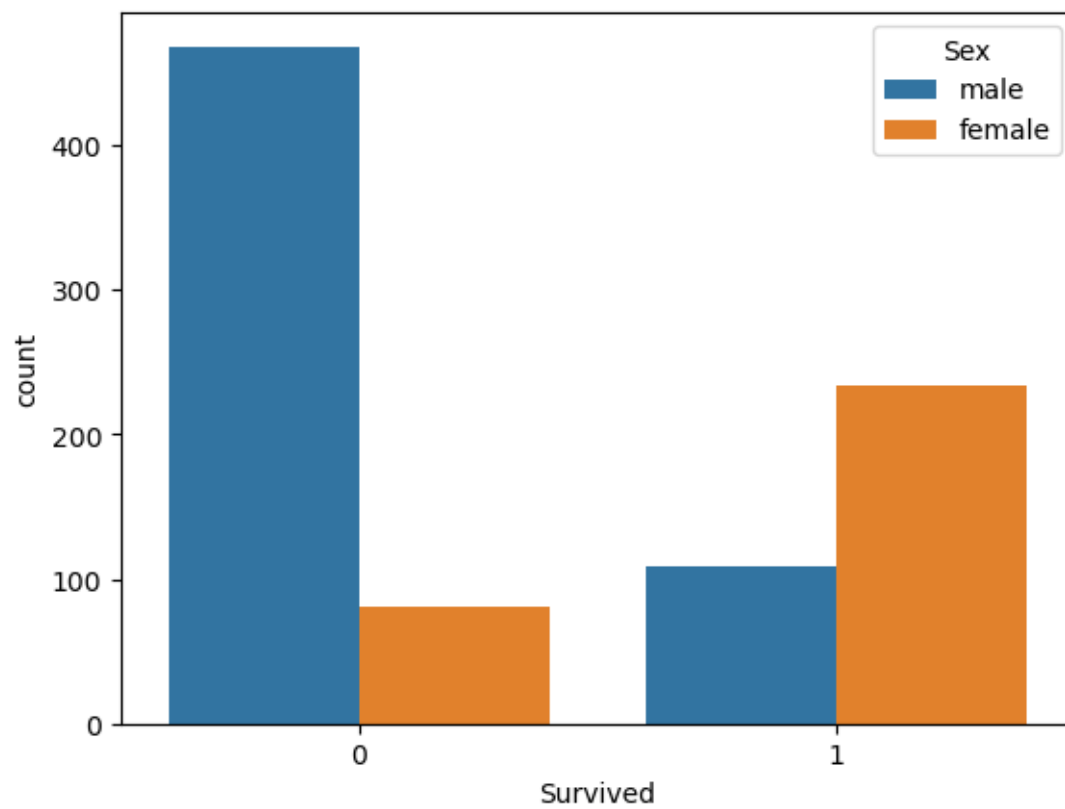
```
[14]: sns.countplot(x='Survived', data=df)
      plt.show()
```



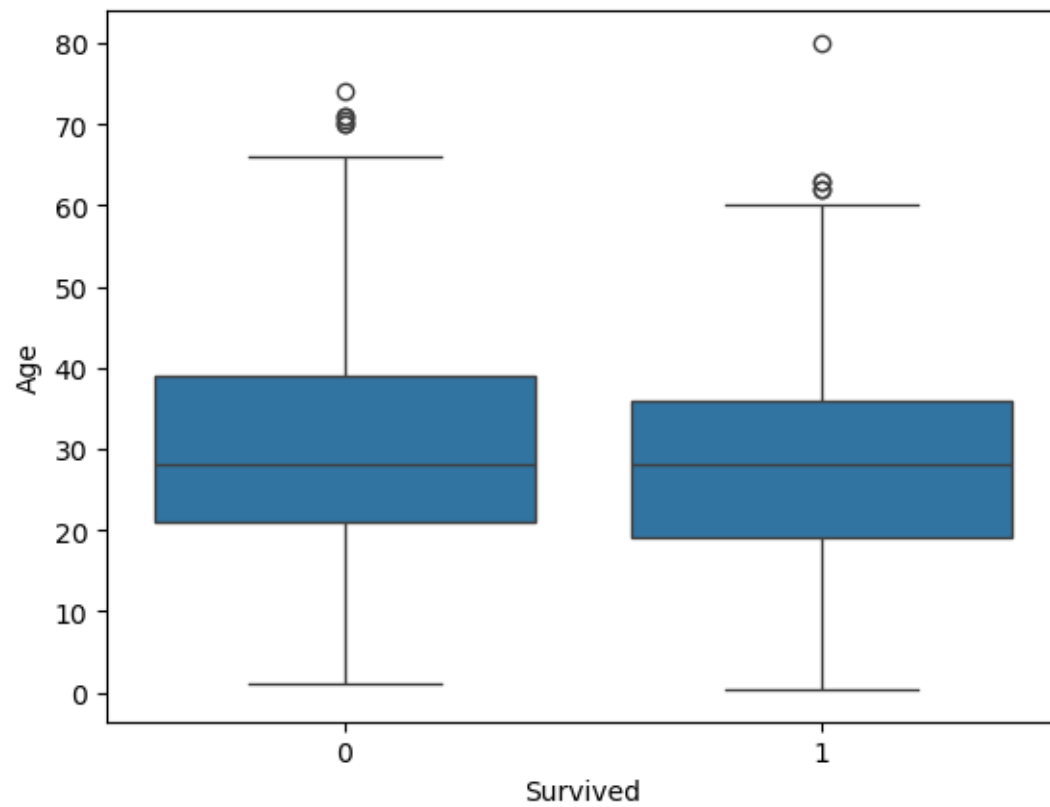
```
[15]: sns.histplot(df['Age'], kde=True)  
plt.show()
```



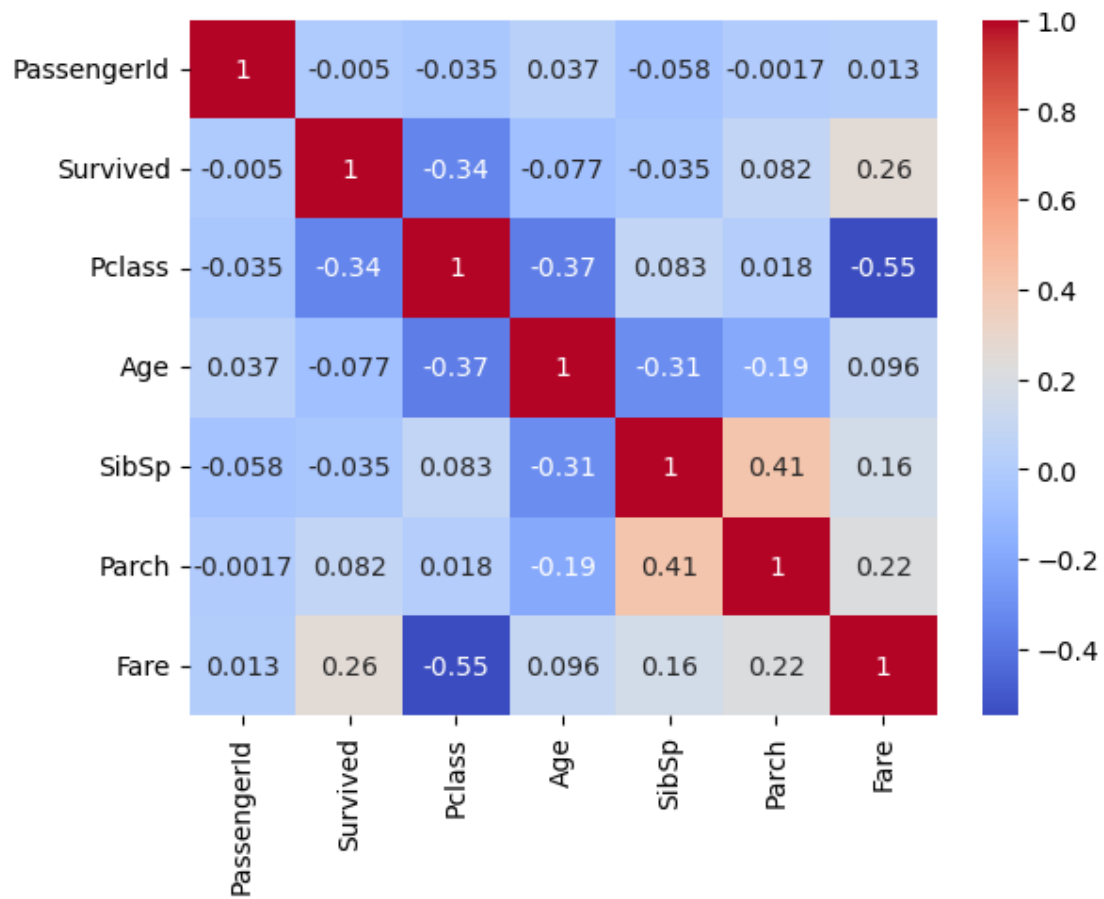
```
[16]: sns.countplot(x='Survived', hue='Sex', data=df)  
plt.show()
```



```
[17]: sns.boxplot(x='Survived', y='Age', data=df)  
plt.show()
```



```
[18]: sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')  
plt.show()
```

[]: